9

SHORT- AND LONG-TERM STABILITY OF CORTICAL ELECTRICAL PHOSPHENES

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A useful visual prosthesis depending on electrical stimulation of the occipital cortex will be possible only if the positions of cortical electrical phosphenes in the visual field are to some extent stable; and the more stable they are, the better.

The blind patient who received the second visual prosthetic implant (Brindley *et al* 1972, Donaldson 1973, Brindley & Rushton 1974) has been intensively tested on nine 'visits' between November 1972 and November 1975. Each 'visit' lasted about a week, and testing occupied about 5 or 6 hours on each day of that week. The present paper will consider the degree of stability of mapping revealed by this testing.

Full maps

Part of each visit was used to make a full map of all the phosphenes then obtainable. The nine maps are shown in Figs. 9.1–9.9 and the layout of the cortical electrodes in Fig. 9.10. The number of phosphenes plotted decreases from map to map because of failures in the extracranial part of the implant.

To make a map, the position of each posphene was fixed by observing its relation to several of its neighbours, and the overall scale and shape of the map was established by making observations between phosphenes in different quadrants. The size and shape of each phosphene is given as described by the subject, and the maps take account of supplementary observations as to whether particular phosphenes touch or overlap each other. Full maps take a day or more to make, and even then it is not practicable to check the relation of every phosphene with every other that is near to it; the relations actually observed will vary from one map to another.

The possibility of making maps and the general agreement between the maps for different visits show that the positions of phosphenes in the visual field are to some extent stable. The differences of detail between maps suggest that the stability is not absolute; they do not prove instability, because they could perhaps be due to errors of mapping. To examine this question further,

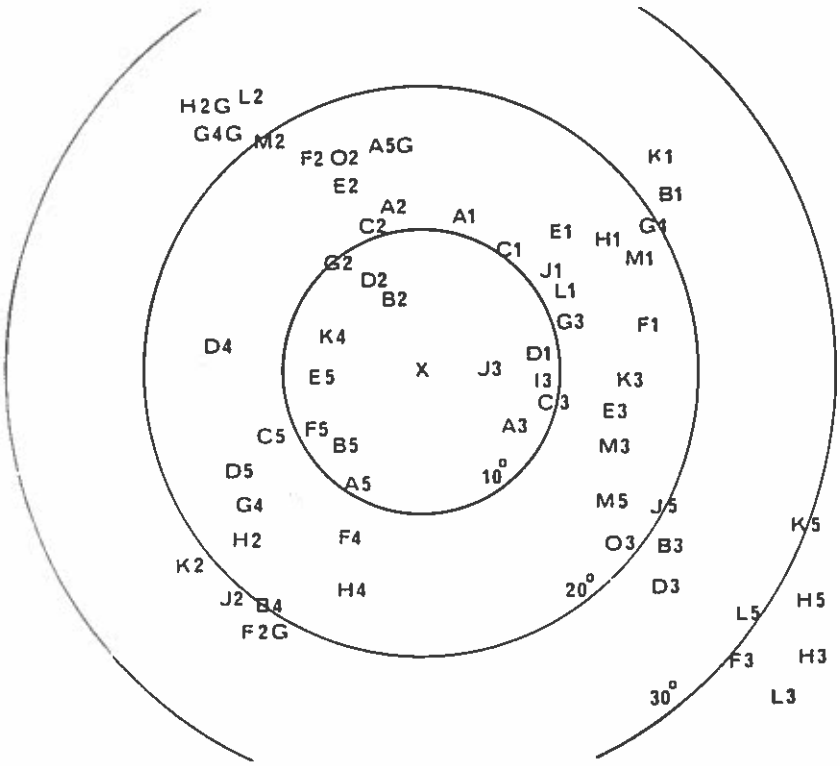

Fig. 9.1 Map of phosphenes made in November 1972. The shapes of the phosphenes on this visit were not carefully recorded, and are not shown. High threshold second phosphenes ('ghosts') are labelled with the suffix G. All maps are shown as seen by the subject, and not as would be plotted to confrontation.

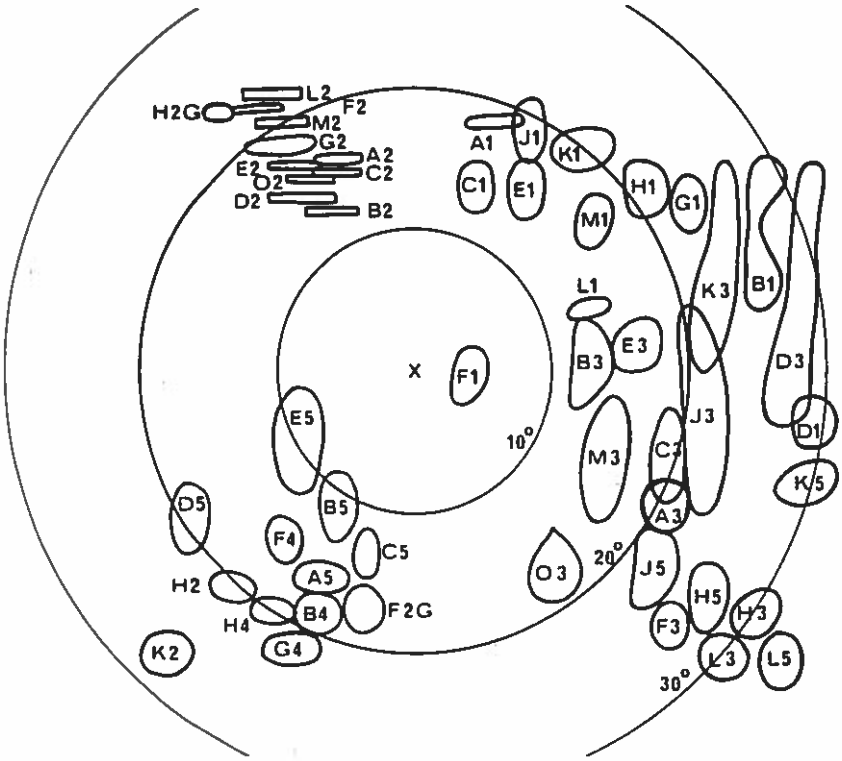

Fig. 9.2 Map of phosphenes made in February 1973. Outlines roughly indicate the size and shape of phosphenes, but many have indistinct edges, and all become larger with strong stimulation.

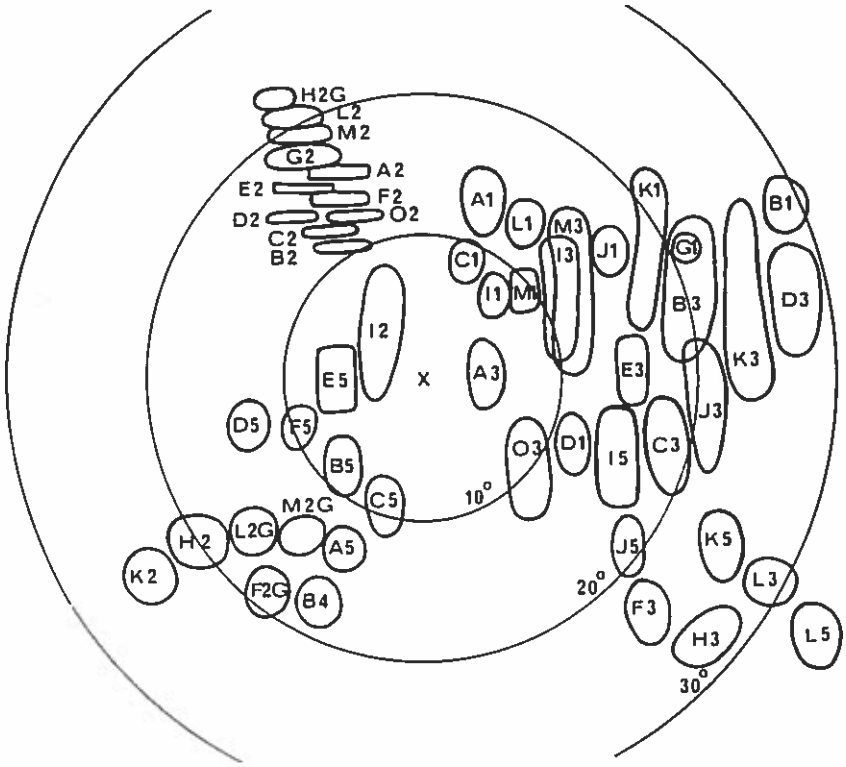

Fig. 9.3 Map of phosphenes made in May 1973.

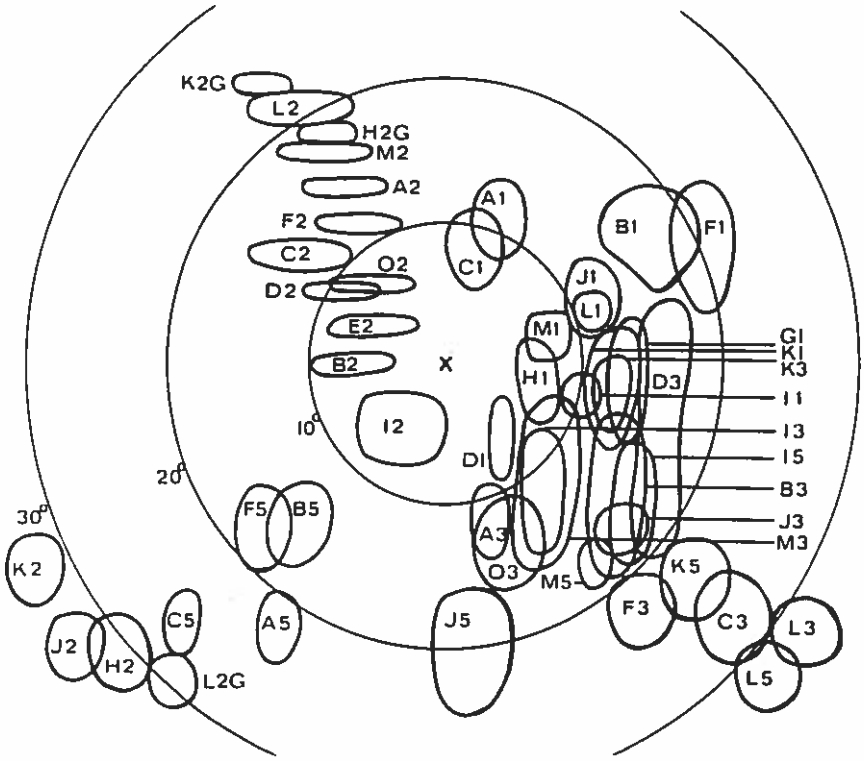

Fig. 9.4 Map of phosphenes made in September 1973.

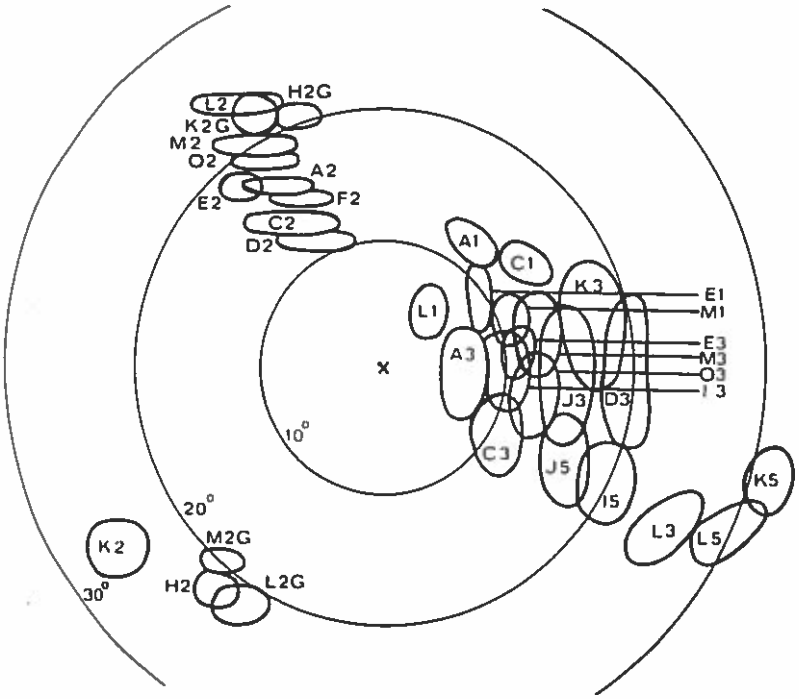

Fig. 9.5 Map of phosphenes made in January 1974.

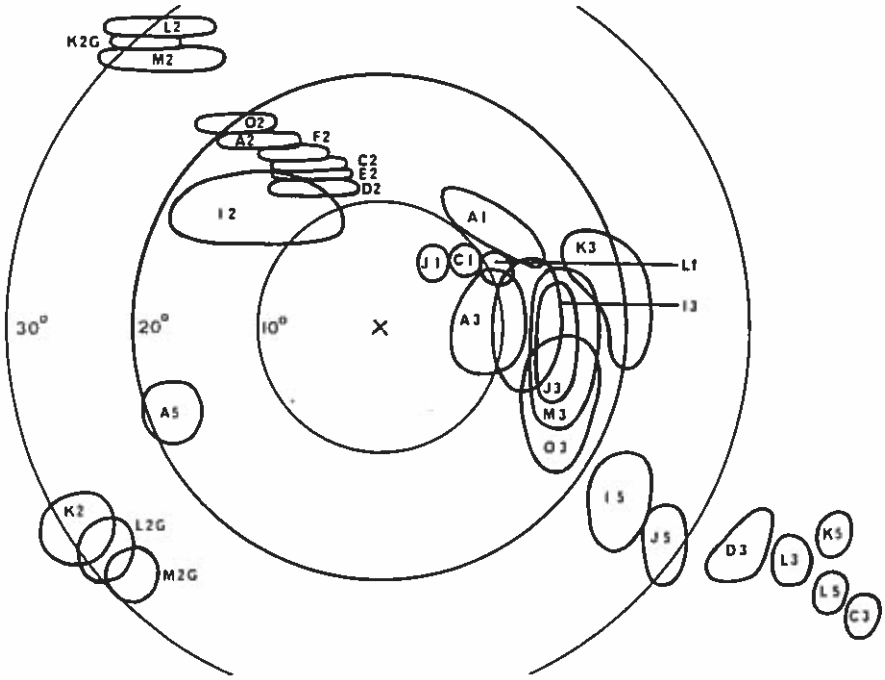

Fig. 9.6 Map of phosphenes made in June 1974.

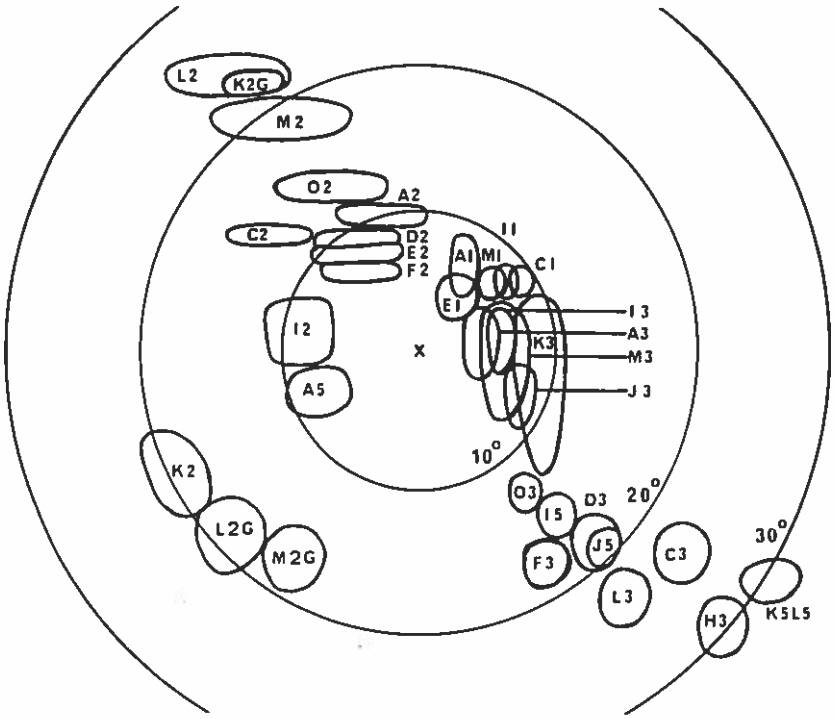

Fig. 9.7 Map of phosphenes made in November 1974.

During this visit, K5 and L5 did not give a phosphene singly, but did when stimulated together.

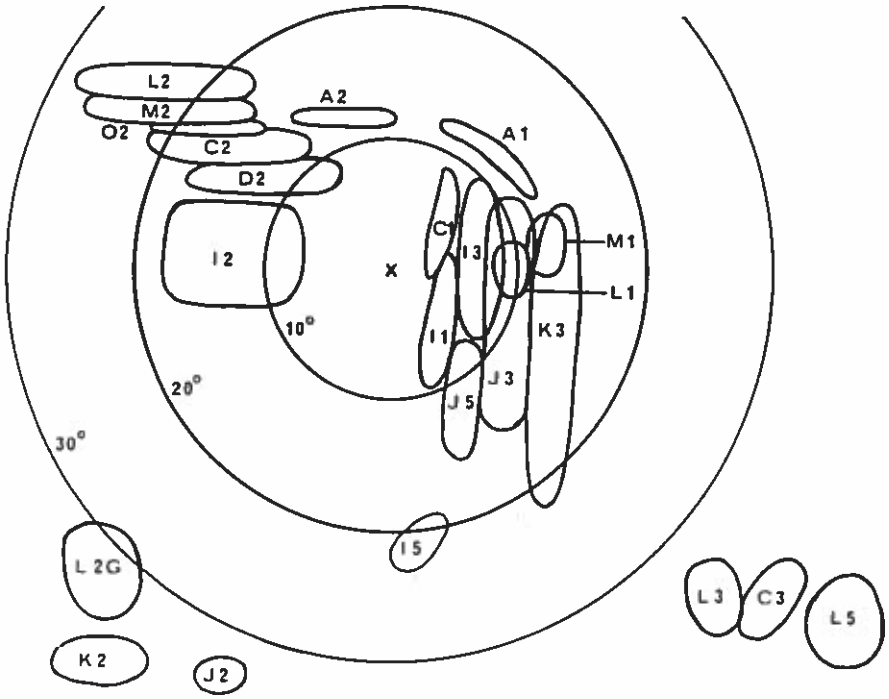

Fig. 9.8 Map of phosphenes made in May 1975.

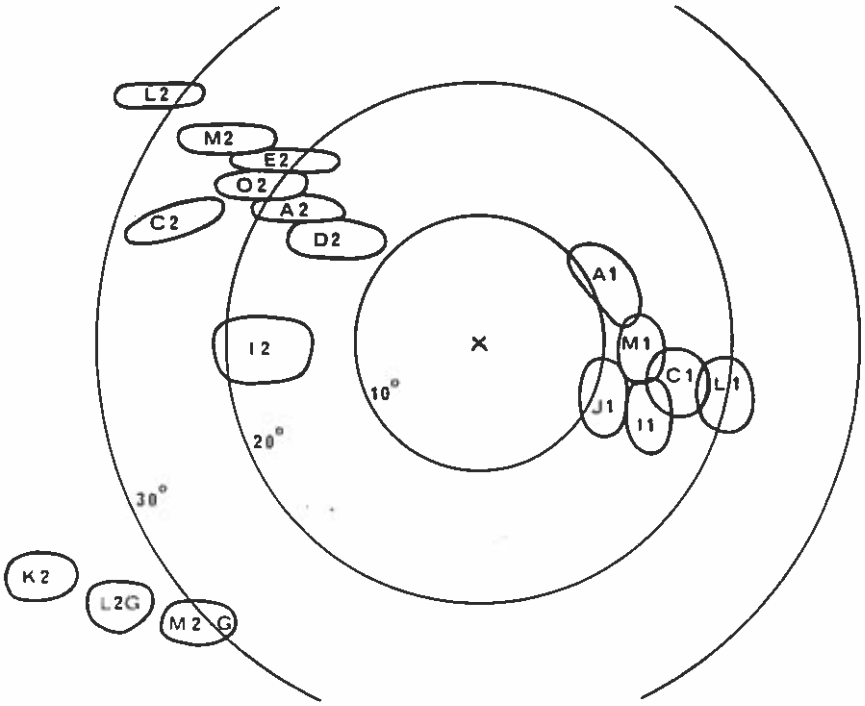

Fig. 9.9 Map of phosphenes made in November 1975.

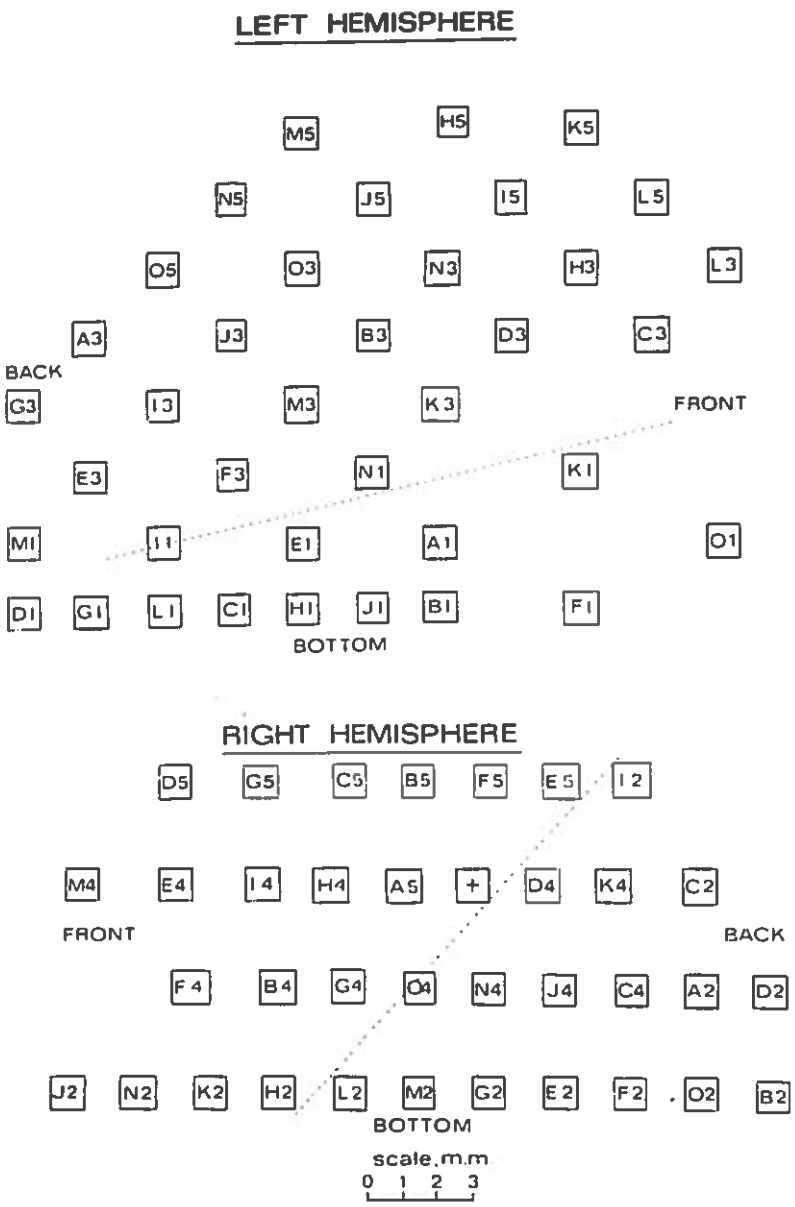

Fig. 9.10 Diagram of arrays of electrodes in contact with medial surfaces and occipital poles of the hemispheres.

we decided to make repeated observations of certain features of the map, and compare the changes over different time intervals. We took a limited set of phosphene pairs ('standard observations'), and repeated observations of their relative position at intervals of minutes, hours, days and months in order to discover the contributions made to changes in the map by real movement of phosphenes, and by the various kinds of error.

Standard observations

In making standard observations, as in making observations in the construction of a full map, the following technique is used. Phosphene A is presented, and is followed after an interval of less than a second by phosphene B. The subject gives the direction and distance of the centre of B from the centre of A. The direction is given as minutes past the hour, 0–59, treating phosphene A as if at the centre of a clock face. The distance is given in inches and fractions of an inch, as would be measured with a ruler at the distance at which the phosphenes are seen. This distance is not always constant, and the effects of changes in it are discussed later.

Standard observations are often made in groups of three, in the form:

Phosphene A→Phosphene B
Phosphene B→Phosphene C
Phosphene C→Phosphene A.

Such a group of three forms a triangle, and a triangle of observations is self checking in that its failure to close gives an estimate of changes or errors operating in the few seconds required to make the observations.

Several such observations are made in a few minutes, and the whole set is then repeated. Pairs of sets of observations are made, up to three times in a day, at intervals of approximately two hours. These observations are repeated daily during a visit by the subject, which usually lasts for about a week, and is repeated every 3–6 months.

Standard observations have been made in nine visits, between November 1972 and November 1975, so that clusters of observations separated by minutes, hours, days and months are available. Short- and longer-term clusters can be compared to discover the time course of any changes in the observed relations of the selected phosphenes. The means and confidence of the means of the positions of phosphenes in clusters of observations can be calculated in order to determine which changes are significant.

Owing to the failure of certain radio receivers in the implant, few of the original standard pairs of phosphenes survived the nine visits. When one phosphene of a triangle failed, a neighbouring phosphene was, where possible, recruited to replace it.

Possible sources of error and variation in mapping and standard observations

Possible sources of difference between maps were listed by Brindley and Rushton (1974), but no attempt was there made to assess the contributions made by real changes in the map, and by errors of different kinds.

Table 9.1 (p. 149) lists some possible sources of internal inconsistency within the observations that comprise a standard triangle, and of differences between successive observations of the same phosphene pairs. The expected character of each kind of variation is given on the right in Table 9.1. The problem is to distinguish real changes in the phosphene map from the other causes of variation. Any real changes in the map are likely to be random, if the map is viewed for a sufficiently long time; phosphenes are unlikely to move in regular paths. Changes may seem, when the map is observed for a shorter time, to be sudden or progressive, or random, depending on the time scale over which they become random. Those changes that occur in a time less than that taken in observing a triangle will give rise to internal inconsistency within that triangle, which will fail to close.

Medium-term changes that are too slow to interfere with the making of a triangle of observations, but are rapid enough to appear random on successive visits (or some shorter interval), will be observed as sources of additional variance occurring over that time interval.

Changes that occur more slowly than the time between visits ('Long-term changes') will appear to be systematic with time, and will be observed as a significant difference between the mean observed relations for successive visits.

Range of data

Twenty-five phosphenes were used in those 42 pairs which survived for standard observations through more than one visit. Many of these pairs were assembled into the triangles discussed above.

Table 9.2 gives, as an example, the 82 observations of the pair C2-L2 made during visits 4 to 8 inclusive.

Table 9.3 shows, for the 7 longest-observed pairs, the overall frequency of observation of different angular relations. Several of the close pairs show a double maximum, either at 20 and 25, or at 50 and 55. The two maxima are scattered throughout all visits, and probably represent a tendency for our subject to give round numbers for the angular relation between phosphenes that are near to each other, rather than being caused by the movement of a phosphene between two alternative positions.

Observed errors and variations in standard observations

A. Internal inconsistency within a triangle of observations

Those sources of variation that act rapidly, during the course of an observation, will cause failure of closure of the triangle of observations in which it occurs. Causes of triangle closure failure can be subdivided into those in which the inconsistency is expected to be repeatable (3b, 4, 5 and 6 in Table 9.1), and those in which the inconsistency is expected to occur at random (1d, 3a and 7 in Table 9.1).

Some of these sources of variation are much more likely than others. Our subject is sure that phosphenes very seldom move during the time required to make an observation (1d) and that a phosphene does not affect the position of a phosphene subsequently given (6). On the other hand, he is aware that seeing a phosphene tends to provoke eye movements directed towards it, and hence to alter the position, on a fixed frame of reference, of a phosphene subsequently given (3). He is also aware that he tends to underestimate large separations in comparison with small ones (5), and that large differences between the apparent distances of two phosphenes from the eye makes estimation of their angular separation difficult (4).

The size and direction of the error by which a triangle of observations fails to close can be represented by a vector, as in Fig. 9.11a, and Fig. 9.11b gives the collection of error vectors obtained in the 82 observations of the longest-observed triangle (L2-D2-C2) through 5 (visits 4–8).

It can be seen that there is a strong tendency towards consistency in the

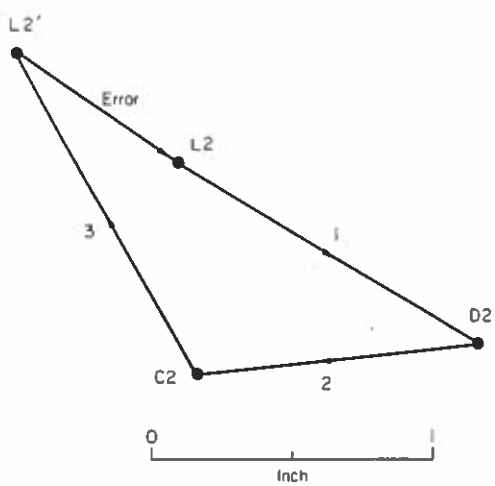

Fig. 9.11a Closure error vector for one observation of the triangle L2–D2–C2. The observations were made in the order shown, from L2 to L2'.

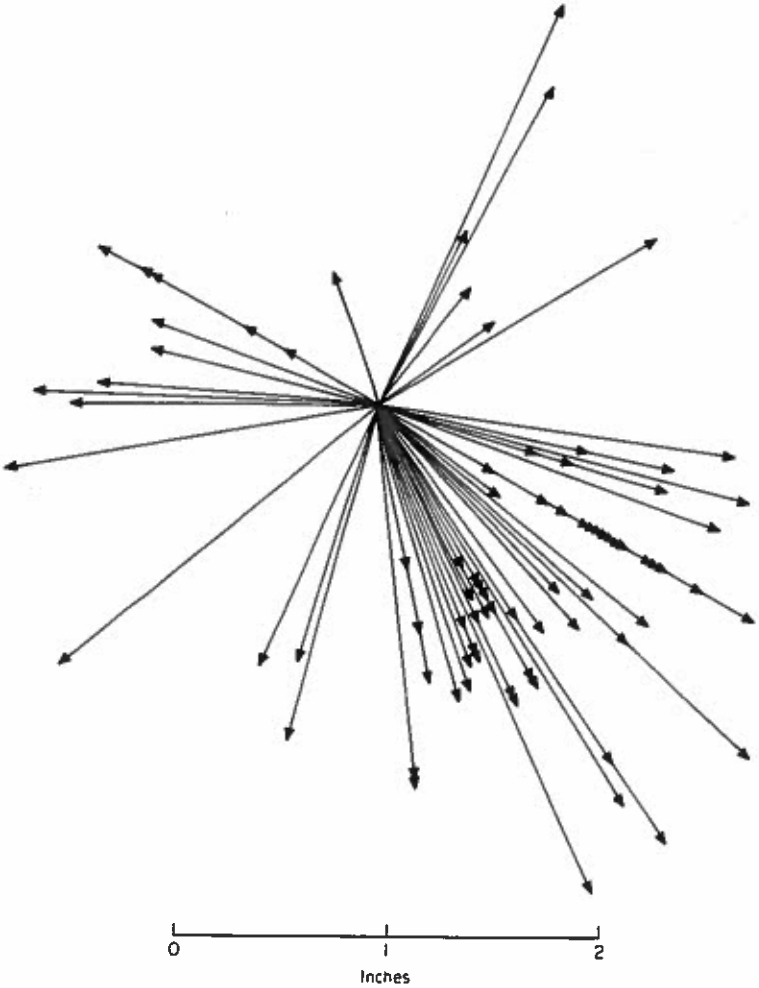

Fig. 9.11b Collection of triangle closure error vectors for the 82 observations of the triangle L2–D2–C2 during visits 4–8. Each arrow head represents the end of one vector.

direction of the error vectors. They tend to lie in the direction of the longest side of the triangle of observations, suggesting either that the underestimation of longer distances is an important source of error in this instance, or that the most peripheral phosphene (L2) has an increased tendency to provoke eye movements towards itself; or both of these.

B. Medium-term variation of standard observations

Phosphenes might move at random about a mean position, in a time much

less than the time between visits. The mean position during a visit would then be unaffected by such movements. The movements could not be large, or they would be detectable as changes in the observed relation from hour to hour, or day to day; but if small they would remain undetected, because too few observations are made in a short time interval for an accurate mean.

Certain phosphenes are particularly likely to *appear* to move in this fashion. The large vertical phosphenes in the right-half field tend to vary from day to day in their shape, and in which part of their area is brightest and appears at lowest current threshold. We ask our subject to make his standard observations between the 'centres of gravity' of two phosphenes, and a change in the shape or distribution of brightness in a phosphene will change his estimate of its 'centre of gravity'.

This kind of random variation is potentially detectable by examining the data for sources of extra variance between observations separated by longer time intervals. However, even where such extra variance is found, it remains possible that the better agreement over short intervals is due to our subject's recollection of the figures he gave for his previous observation. The reason for this is that, although we do not tell our subject which of his standard observations is to be presented next, his phosphenes are so familiar to him that he can nevertheless often recognize them. This is likely to be the explanation for any extra variance added by an interval of hours.

Table 9.4 shows, for the seven longest-observed phosphene pairs, which time intervals were sources of significant extra variance. Extra variance introduced merely by changes in separation and not angle is marked by a letter D, and may represent changes in the scale of the map. Changes in the scale of the map are likely in themselves to cause changes in the direction of observations involving K2 and perhaps other low phosphenes, for reasons that are discussed later. After these exceptions are allowed for, the two long-observed pairs that show extra variance over days or months both include C2 (C2-L2 and D2-C2).

C. Long-term variation of standard observations

Long-term variation in a standard observation pair will be seen to have occurred when a consistently obtained relation gives place to a different relation that is also consistent, or when a significant progressive change occurs. In Table 9.1, hypotheses (1a), (1b), (2) and (8) are expected to give changes of this character. The effects of (8) are discussed later. In (1) the phosphenes may move singly, while in (2) all the phosphenes would be expected to move, although not equally, owing to variations in the cortical magnification factor (Daniel & Whitteridge, 1961).

We do not believe that such movement of the electrode array over the cortex ever actually occurred, after the first few days, because the electrode

array would be expected to have become encapsulated by secondary dura (Brindley & Lewin 1968, Craggs & Rushton 1976). There has been no visible shift of the electrode arrays on X-rays taken during the 4 years since implantation.

Visit means

We looked for long-term changes by calculating the mean angle and separation between each standard pair for each visit. Table 9.5 gives the means for the seven longest-observed phosphene pairs, and the number of observations on which each mean is based. In order to determine whether the changes are significant, we calculated, for each visit mean, the size and shape of the ellipse of 95% confidence in the mean position of the second phosphene of each pair, arbitrarily taking the position of the first as being fixed. The ellipses were calculated on the assumption of bivariate normality of the observations. The overlap of the 95% confident ellipses around the visit means is given for the seven long-observed pairs in Fig. 9.12, a–g.

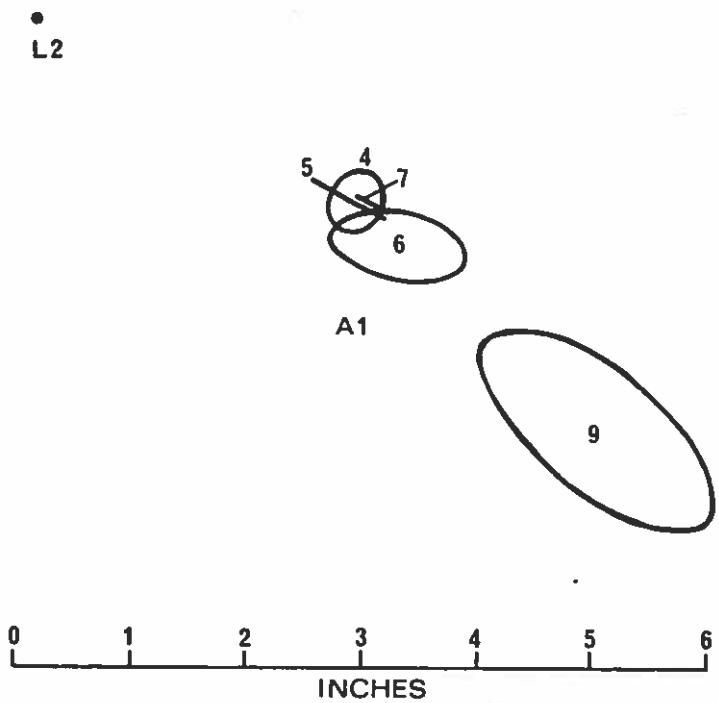

Fig. 9.12a Ellipses of 95% confidence in the mean relation L2–A1, for each of visits 4, 5, 6, 7 and 9. L2 is arbitrarily taken as fixed for this purpose, but turning the figure upside down shows the result where A1 is taken as fixed. For visits 5 and 7, the ellipse is of zero width, since the observed angle was unvaried throughout these visits.

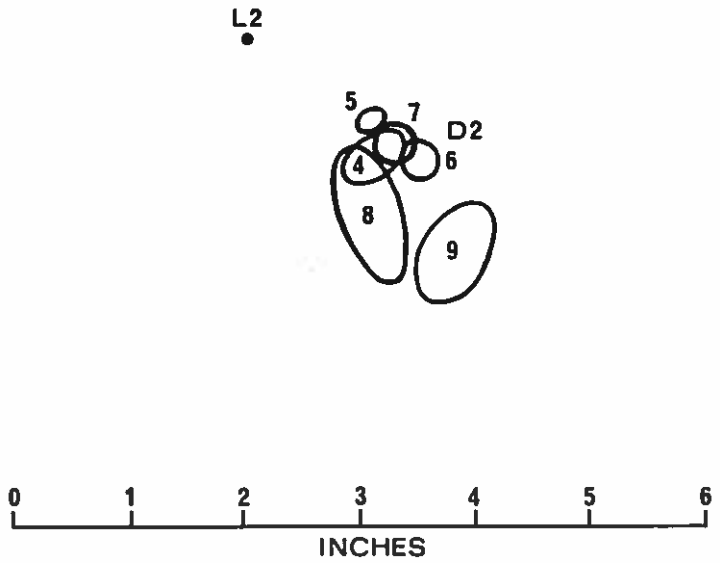

Fig. 9.12b Ellipses of 95% confidence in the mean relation L2–D2, for each of visits 4, 5, 6, 7, 8 and 9.

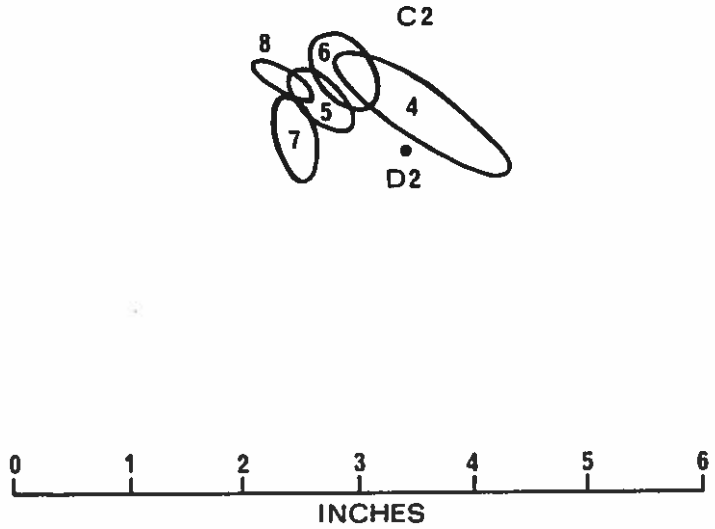

Fig. 9.12c Ellipses of 95% confidence in the mean relation D2–C2, for each of visits 4, 5, 6, 7 and 8.

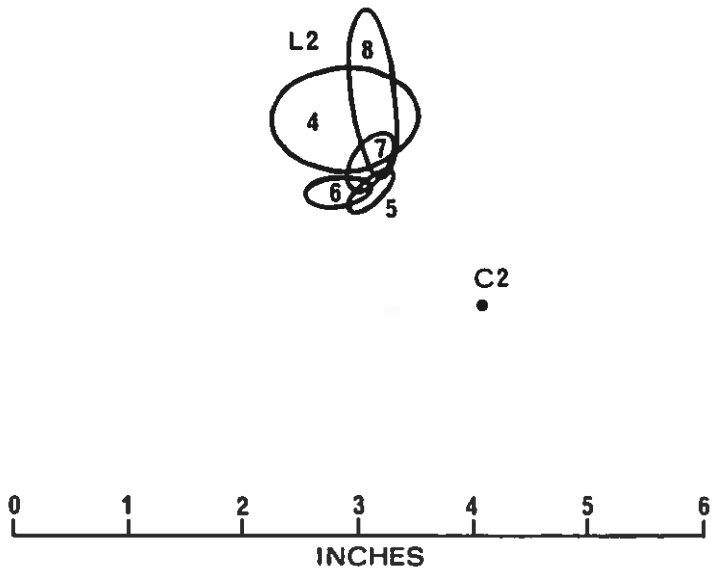

Fig. 9.12d Ellipses of 95% confidence in the mean relation C2–L2, for each of visits 4, 5, 6, 7 and 8.

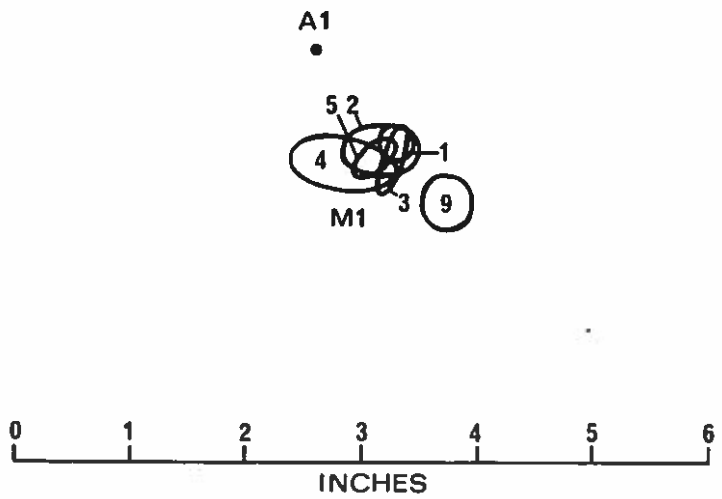

Fig. 9.12e Ellipses of 95% confidence in the mean relation A1–M2, for each of visits 1, 2, 3, 4, 5 and 9.

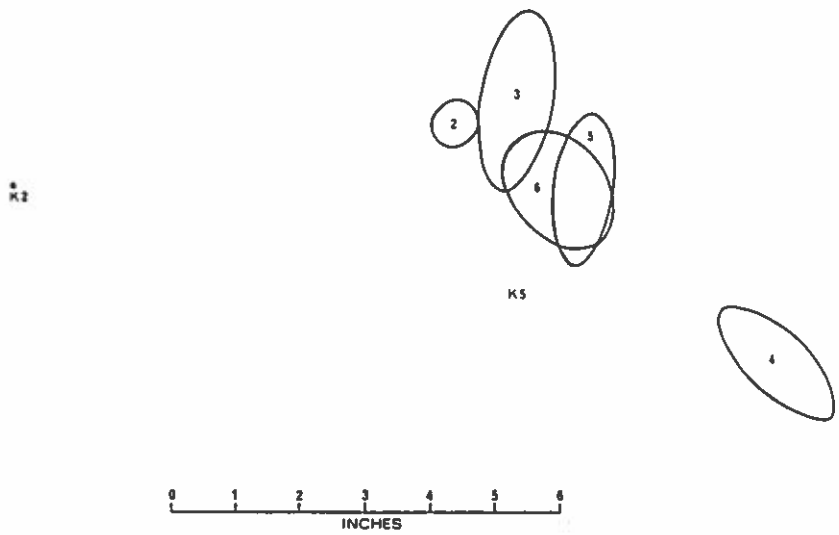

Fig. 9.12f Ellipses of 95% confidence in the mean relation K2–K5, for each of visits 2, 3, 4, 5 and 6. There were insufficient observations in visit 1 to calculate the confidence of the mean value.

We expected observations of angle to be more reliable than observations of separation, both within and between visits. We therefore expected that visit mean ellipses would in general have their long axes along a line joining the two phosphenes, and that the various visit mean ellipses would be spread out mainly along the same line. Both of these expectations were fulfilled in the case of L2-A1, and the latter for L2-D2 and A1-M1.

Visit means for C2-L2, A1-M1 and L2-D2 were very consistent. D2-C2 is the only short-distance pair that appears unstable, the mean for visit 4 being different from that for later visits. As may be judged from the size of the ellipse for visit 4, its observations were then highly variable (and rather few). A suspicion that D2-C2 may have changed is compatible with their rather variable relation on the full maps, and with the analysis shown in Table 9.4.

The three long-distance observations L2-A1, K2-K5, K2-L2, all varied significantly, the first in separation only, and the others in both angle and separation. The variation in direction of observations involving K2 is probably attributable to the great and variable difference between the distance of K2 from the eye and the distances of the other phosphenes.

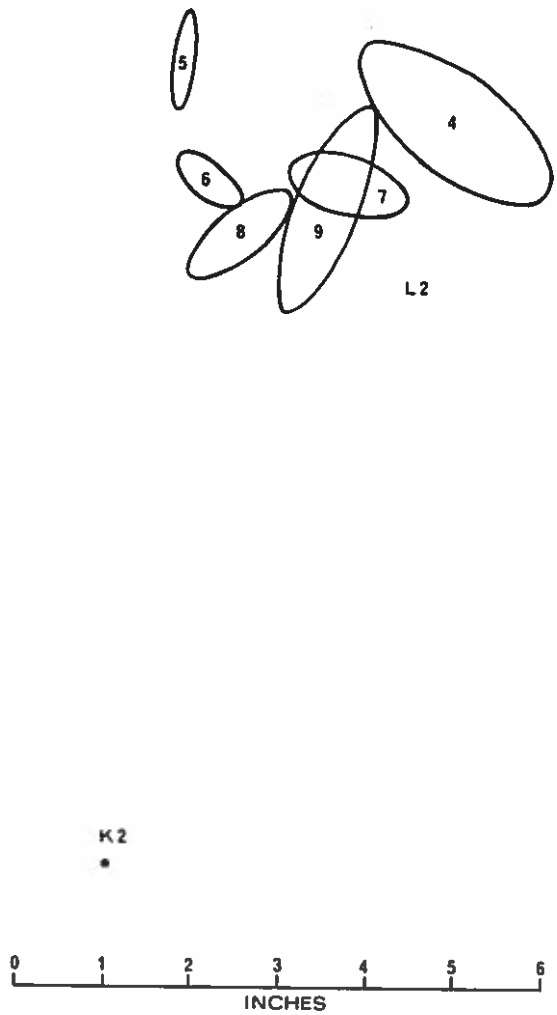

Fig. 9.12g Ellipses of 95% confidence in the mean relation K2–L2 for each of visits 4, 5, 6, 7, 8 and 9.

Variations in the scale of the map and the apparent distance of phosphenes

A tendency for visit means to vary more in separation than in angle has been noted above. If this variation is caused by changes in the overall scale of the map (which would be most plausibly explained by changes in the apparent distance of the phosphenes from the observer), then all phosphene separations should vary in proportion. This is not always the case, as is shown by Table 9.5, which gives the mean separation for each visit for each of the seven longest-observed phosphene pairs. Although there is a tendency for the separations to increase in later visits, this does not apply to all phosphene pairs.

The explanation for this may be that our subject does not see all phosphenes as being at the same distance, or as varying in distance in the same way. In particular, the lowest phosphenes are seen as being much closer than the higher phosphenes. This is shown in Fig. 9.13 for the most recent

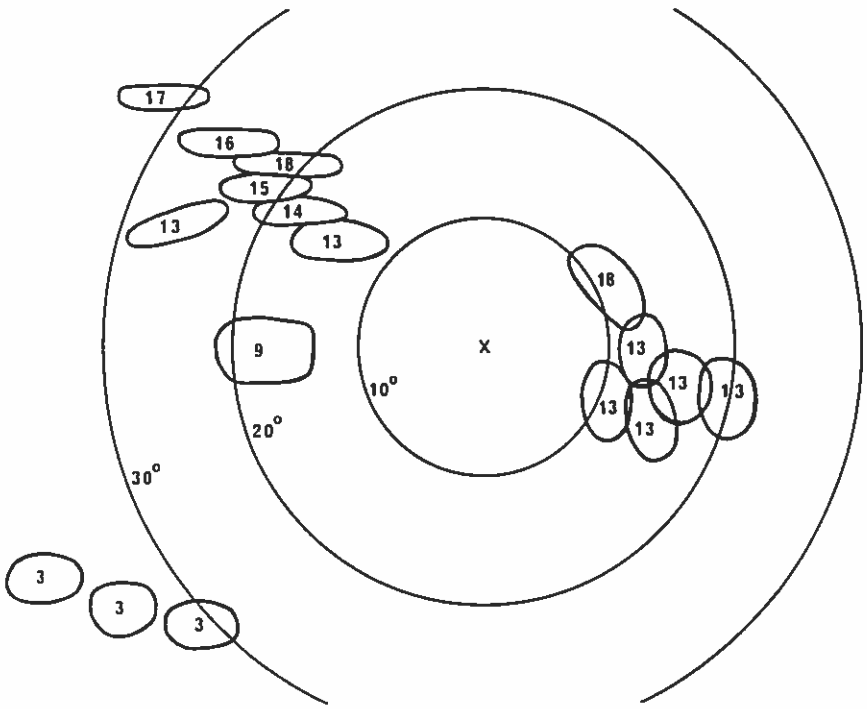

Fig. 9.13 Map of phosphenes made in November 1975. The identity of the phosphenes is as given in Fig. 9.9, but they are labelled with their stated apparent distance from the eye in inches, as observed by the subject.

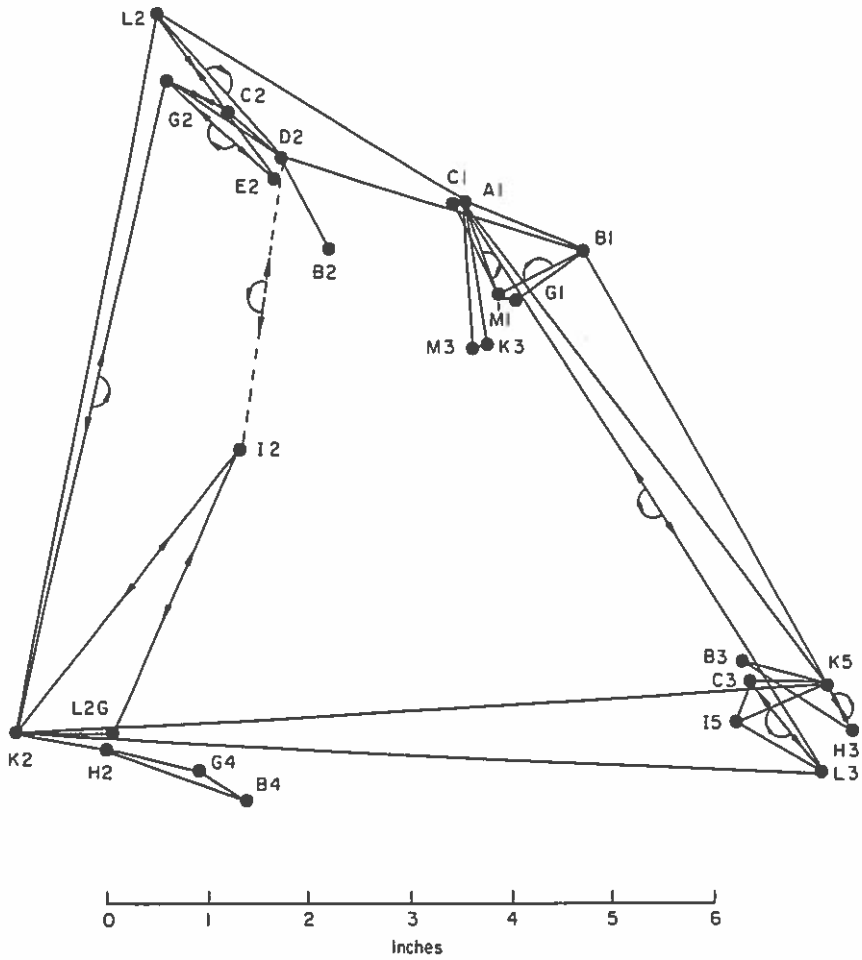

Fig. 9.14 Map obtained by triangulation of the mean relations of those 42 standard observation pairs (25 phosphenes) that were observed many times during more than one visit. Distortion introduced in the fitting, of more than 10° (in angle) or more than 10% (in separation), is shown for the relations in which it was necessary.

indicates that the angle had to be changed in a clockwise direction, in order to fit; and vice versa.

indicates that the separation had to be reduced, in order to fit; and vice versa.

The relation D2–I2, which was never a standard observation, but which has been observed on 11 occasions, is dotted in, since it fills an important gap in the skeleton map.

map (November 1975, visit 9), which is the only visit in which the distances of phosphenes from the eye were carefully recorded.

Compatibility between the means for the different standard observations

If the phosphene map is considered to be stable, subject to the exceptions discussed above, then it is permissible to calculate a grand mean for all observations of each standard pair. These pairs can then be fitted together into a many-times observed skeleton of a map. The random, but not the systematic, causes of error listed in Table 9.1 will be reduced in this way, and the goodness of fit of the parts of the skeleton will give an indication of the size of the systematic causes of error.

The result of this procedure applied to those 42 standard pairs (25 phosphenes) that were observed during more than one visit is shown in Fig. 9.14. Certain distortions of angle and separation had to be introduced, in order to make the skeleton fit together into a two dimensional map. Where the distortions were larger than a 10° angle or 10% of separation, the way in which they had to be distorted is shown.

Conclusion

Most of our subject's phosphenes have remained in the same part of the field throughout the time that we have studied them. The main exception to this general statement is K2, which moved from a very high to a very low position in the left field during the last week of February 1972, three weeks after implantation. A high threshold second phosphene remained in the original position, and it seems likely that the change was due to a slight early shift of the electrode array, which happened to move the electrode K2 from the lower to the upper lip of the calcarine fissure.

Certain phosphenes seem to be unstable, and to move about at random within a small area. Of the 42 phosphene pairs that were repeatedly observed over more than one visit, ten showed a signficant extra variation in direction that was added by an interval of days, while two of these, and nine others, showed a significant extra variation in direction that was added by an interval of months.

Although significant, none of these changes was very great, and the largest of them are included in the longest-observed pairs shown in Fig. 9.12, and Tables 9.4 and 9.5.

Twenty-three pairs gave no significant extra variation of direction with intervals of days or months, when compared with shorter intervals, and were, as far as we could tell, perfectly stable.

Acknowledgments

We are grateful to our subject, F.R.B., for his patient cooperation and careful observations throughout many hundreds of hours of testing; and to D. G. Clayton for statistical advice and for performing the bivariate analyses.

Table 9.1 Left column lists some possible causes of differences between successive observations of a triangle of cortical phosphenes, and of internal inconsistencies within such triangles of observations. On the right are shown the expected features of the change or inconsistency

	Source of difference between maps	Expected result in Standard observations
1.	Real changes in the map	
	a. Slow, progressive (long term)	Systematic drifts in observed relation
	b. Rare, sudden (long term)	Stepwise changes in observed relation
	c. Random (medium term)	Random variation in observed relation
	d. Random (short term)	Random failure of triangle closure
2.	Movement of electrode array	Systematic change affecting all phosphenes
3.	Eye movements	
	a. Random	Random failure of triangle closure
	b. Phosphene-provoked	Systematic failure of triangle closure
4.	Phosphenes appearing to be at different distances from the eye	Systematic failure of triangle closure
5.	Errors of judgement of small as against large separations	Systematic failure of triangle closure, where the sides of triangles are unequal
6.	Phosphene A affecting the subsequent position of Phosphene B	Systematic failure of triangle closure
7.	Random errors of judgement of angle or separation	Random variation of observed relations, and random failure of triangle closure
8.	Changes in the apparent distance of phosphenes from the eye	Change in the observed separation of phosphene pairs, without necessarily any change in their angular relationship

Table 9.2 The table lists all 82 observations of the standard pair C2-L2, made during visits 4–8. Observations separated by minutes are placed together, and intervals of hours are represented by a space

	Visit				
Day	4	5	6	7	8
1	53′, 2″ 55′, 1¼″ 50′, 1″ 59′, 2″	55′, 1¼″ 57′, 1½″ 55′, 1½″ 58′, 1½″	50′, 1¼ 50′, 1½″	55′, 2″ 52′, 2″	55′, 1″ 55′, 1½″ 50′, 1½″ 53′, 1½″
2	55′, 1½″ 50′, 2″ 50′, 3″ 55′, 3″	50′, 1½″ 50′, 2″ 55′, 1½″ 50′, 1½″	50′, 1½″ 55′, 1¼″ 50′, 1½″ 50′, 1½″	50′, 2¼″ 55′, 1½ 55′, 2″ 58′, 1¼″	53′, 1¼″ 55′, 2″
3		55′, 1½″ 50′, 1½″ 50′, 1½″ 55′, 1½″ 50′, 1¼″	53′, 1½″ 55′, 1½″ 50′, 1¼″ 55′, 1¼″ 50′, 1½″ 50′, 2″	57′, 2″ 55′, 1¼″	55′, 2½″
4		53′, 1½″ 50′, 1½″ 50″, 1¼″ 50′, 1½″	55′, 2″ 50′, 2″ 50′ 2½″ 55′, 1½″ 59′, 1¼″ 53′, 1¾″	55′, 1¼″ 55′, 1¼″ 50′, 1¼ 55′, 1¾″ 50′, 1½″ 55′, 1¾	55′, 1¼″ 56′, 3″ 57′, 3½″ 57′, 3½″
5		53′, 1½″ 50′, 1½″	50′, 1½″ 50′, 1½″	50′, 1¼″ 55′, 1¼″ 50′, 1¼″ 55′, 1¼″	
6			50′, 1½″ 50′, 2″ 50′, 2″ 50′, 2½″ 50′, 2½″ 50′, 2¼″		

Table 9.3 The table shows the overall frequency of observation of different angular relations, for the seven longest-observed standard phosphene pairs. The angle is shown in minutes past the hour, as observed by the subject, treating the first phosphene of each pair as if at the centre of a clock face, and the second as at the edge

Minutes	L2–A1	L2–D2	D2–C2	C2–L2	A1–M1	K2–K5	K2–L2
00							1
01							39
02							19
03			1				24
04							4
05							16
07							1
08							1
10						4	
12						7	
13						25	
14					3	16	
15						2	
16			1			9	
17			1			14	
18					1	4	
20	89	68	5		24		
21					1		
22					3		
23		14			11		
24					2		
25	6	17	1		29		
26		1			4		
27	1	3			5		
28					5		
29		1			5		
30					2		
31			1		1		
35			1				
39	1						
40			6		2		
43			1		1		
44			1		1		
47			1				
48			2				
50			38	37			
52				1			
53			3	7			
55			17	28			
56				1			
57				4			
58				2			
59			2	2			
Total	97	104	82	82	100	81	105

Table 9.4 The table gives the mean, for each visit, of the angle and separation of the seven longest-observed phosphene pairs. The angle from the first to the second is given as minutes past the hour, 0–59, to one decimal. The mean separation is given in inches and decimals, to three significant figures. The number of observations contributing to each mean is given. The right-hand column gives the overall bivariate means, for each of the seven pairs

	Visit									
Pair	1	2	3	4	5	6	7	8	9	Overall mean
L2-A1				20·0′ 3·19″ N = 8	20·0′ 3·00″ N = 19	20·6′ 3·65″ N = 26	20·0′ 3·25″ N = 18		21·1′ 5·95″ N = 26	20·6′ 4·06″ N = 97
L2-D2				21·6′ 1·49″ N = 8	20·6′ 1·28″ N = 19	20·8′ 1·82″ N = 26	20·9′ 1·58″ N = 18	24·1′ 1·88″ N = 11	22·8′ 2·62″ N = 22	21·8′ 1·82″ N = 104

D2-C2				03·4′ 0·33″ N = 8	50·3′ 0·89″ N = 19	53·6′ 0.87″ N = 26	46·2′ 0·97″ N = 18	49·8′ 1·25″ N = 11		50·6′ 0·83″ N = 82
C2-L2				53·7′ 2·00″ N = 8	52·4′ 1·42″ N = 19	51·3′ 1·67″ N = 26	53·6′ 1·65″ N = 18	55·1′ 2·05″ N = 11		52·9′ 1·67″ N = 82
A1-M1	23·0′ 1·03″ N = 15	24·3′ 1·02″ N = 28	23·3′ 1·18″ N = 5	27·4′ 0·97″ N = 8	25·2′ 1·04″ N = 19				23·1′ 1·71″ N = 25	24·3′ 1·32″ N = 100
K2-K5	10·0′ 13·5″ N = 2	13·4′ 7·32″ N = 28	13·2′ 8·34″ N = 8	17·2′ 13·1″ N = 8	14·8′ 9·39″ N = 19	14·9′ 8·99″ N = 16				14·4′ 8·85″ N = 81
K2-L2				04·2′ 9·52″ N = 8	01·0′ 9·36″ N = 19	01·5′ 7·99″ N = 26	03·4′ 8·33″ N = 18	02·1′ 7·48″ N = 8	03·2′ 7·96″ N = 26	02·4′ 8·31″ N = 105

Table 9.5 The table shows, for the seven longest-observed phosphene pairs, which time intervals add significant extra variance to the observations, when compared with the next shorter time intervals. The observations were subjected to a hierarchical bivariate analysis of variance, assuming bivariate normality of the observations. The probability that any extra variance is attributable to chance was calculated.

P 0·05 represented as*
P 0·01 represented as***

Where the extra variance was attributable to changes in the apparent separation, and not the angle between phosphenes, it may be due merely to changes in the scale of the map. The suffix D (for distance) is added in these cases

Pair	Interval		
	Hours	Days	Months
L2-A1		***D	
L2-D2	***	*D	***D
D2-C2	***D		***
C2-L2		*	
A1-M1	***		
K2-K5	***		***
K2-L2	***D	***	***

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